

pyGDIS CSV Example Guide

Purpose

This guide explains how to run the CSV example included with pyGDIS and how to interpret its outputs. The example demonstrates that GDIS can analyze trajectory data stored in a CSV file without requiring knowledge of the governing equations.

Required files

Place the following files in the same directory:

- csv_data_example.py
- synthetic_parameter_sweep.csv

Run the example

```
python csv_data_example.py
```

Console output

```
Input file: synthetic_parameter_sweep.csv
Rows: 7500
Parameter values: 25
State columns: x1, x2
Resolved critical value: 0.541667
Critical-value source: data_driven_transition_energy_peak
Maximum GDIS: 0.852530 at parameter 0.583333
Maximum transition energy at parameter: 0.541667
Output folder: gdis_results/
```

Meaning of the output

Rows: Total number of observations read from the CSV.

Parameter values: Number of unique operating conditions (one trajectory per parameter value).

State columns: Variables used to compute the GDIS descriptors.

Resolved critical value: Estimated parameter where the major dynamical transition occurs.

Critical-value source: Method used to determine the critical point. Here it is estimated automatically from the peak of the transition-energy functional.

Maximum GDIS: Highest instability score observed. Larger values indicate stronger instability.

Maximum transition energy: Parameter where the localized transition dynamics are strongest. This often occurs slightly before the maximum GDIS because the transition precedes fully developed instability.

Output folder: Directory containing all generated results.

Generated files

- **gdis_results.csv**: GDIS scores and descriptor values for every parameter.
- **gdis_profile.png**: GDIS versus parameter.
- **gdis_components.png**: Individual descriptor and component contributions.
- **analysis_summary.txt**: Plain-text summary of the analysis.

Typical workflow

1. Export trajectory data to CSV.
2. Specify the parameter, time, and state columns.
3. Run the example script.
4. Review the generated figures and CSV table.
5. Use the GDIS profile to identify stable, transitional, and highly unstable operating regions.

Using your own data

```
python csv_data_example.py --input my_data.csv --parameter-column load
--time-column time --state-columns voltage angle
```